

This folder contains the [source code of the lean and mean solver from John Tromp for Cuckatoo](#). The source was only slightly modified to have the same header length for the mean solver ("#define HEADERLEN 80" -> "#define HEADERLEN 246" in line 751 of the "mean.cu" file and accordingly in line 265 of the "lean.cu" file). The [Makefile](#) can be used to create and run the binary files to reproduce the evaluation presented [here](#) for claiming the [bounty offered by John Tromp](#).

List of all binaries and their parameter used for the evaluation

Exec. File	Output File	Variant	Seed	Nonces	Threads	k	Memory used	Batches
splean27x8_k10_t1	../eval/data/splean27x8_k10_t1_seed0.out	Cuckatoo27	00000000000000000000...0000	0-3	1	10	1.6 MiB	4
splean27x8_k10_t1	../eval/data/splean27x8_k10_t1_seed1.out	Cuckatoo27	000005eed10000000000...0000	0-3	1	10	1.6 MiB	4
splean27x8_k10_t1	../eval/data/splean27x8_k10_t1_seed2.out	Cuckatoo27	000000000005eed200000...0000	0-3	1	10	1.6 MiB	4
splean27x8_k10_t2	../eval/data/splean27x8_k10_t2_seed0.out	Cuckatoo27	00000000000000000000...0000	0-3	2	10	1.6 MiB	4
splean27x8_k10_t2	../eval/data/splean27x8_k10_t2_seed1.out	Cuckatoo27	000005eed10000000000...0000	0-3	2	10	1.6 MiB	4
splean27x8_k10_t2	../eval/data/splean27x8_k10_t2_seed2.out	Cuckatoo27	000000000005eed200000...0000	0-3	2	10	1.6 MiB	4
splean27x8_k10_t4	../eval/data/splean27x8_k10_t4_seed0.out	Cuckatoo27	00000000000000000000...0000	0-3	4	10	1.6 MiB	4
splean27x8_k10_t4	../eval/data/splean27x8_k10_t4_seed1.out	Cuckatoo27	000005eed10000000000...0000	0-3	4	10	1.6 MiB	4
splean27x8_k10_t4	../eval/data/splean27x8_k10_t4_seed2.out	Cuckatoo27	000000000005eed200000...0000	0-3	4	10	1.6 MiB	4
splean27x8_k10_t8	../eval/data/splean27x8_k10_t8_seed0.out	Cuckatoo27	00000000000000000000...0000	0-3	8	10	1.6 MiB	4
splean27x8_k10_t8	../eval/data/splean27x8_k10_t8_seed1.out	Cuckatoo27	000005eed10000000000...0000	0-3	8	10	1.6 MiB	4
splean27x8_k10_t8	../eval/data/splean27x8_k10_t8_seed2.out	Cuckatoo27	000000000005eed200000...0000	0-3	8	10	1.6 MiB	4
splean27x8_k11_t1	../eval/data/splean27x8_k11_t1_seed0.out	Cuckatoo27	00000000000000000000...0000	0-3	1	11	842.1 KiB	4
splean27x8_k11_t1	../eval/data/splean27x8_k11_t1_seed1.out	Cuckatoo27	000005eed10000000000...0000	0-3	1	11	842.1 KiB	4
splean27x8_k11_t1	../eval/data/splean27x8_k11_t1_seed2.out	Cuckatoo27	000000000005eed200000...0000	0-3	1	11	842.1 KiB	4
splean27x8_k11_t2	../eval/data/splean27x8_k11_t2_seed0.out	Cuckatoo27	00000000000000000000...0000	0-3	2	11	844.1 KiB	4
splean27x8_k11_t2	../eval/data/splean27x8_k11_t2_seed1.out	Cuckatoo27	000005eed10000000000...0000	0-3	2	11	844.1 KiB	4
splean27x8_k11_t2	../eval/data/splean27x8_k11_t2_seed2.out	Cuckatoo27	000000000005eed200000...0000	0-3	2	11	844.1 KiB	4
splean27x8_k11_t4	../eval/data/splean27x8_k11_t4_seed0.out	Cuckatoo27	00000000000000000000...0000	0-3	4	11	844.2 KiB	4
splean27x8_k11_t4	../eval/data/splean27x8_k11_t4_seed1.out	Cuckatoo27	000005eed10000000000...0000	0-3	4	11	844.2 KiB	4
splean27x8_k11_t4	../eval/data/splean27x8_k11_t4_seed2.out	Cuckatoo27	000000000005eed200000...0000	0-3	4	11	844.2 KiB	4
splean27x8_k11_t8	../eval/data/splean27x8_k11_t8_seed0.out	Cuckatoo27	00000000000000000000...0000	0-3	8	11	844.2 KiB	4
splean27x8_k11_t8	../eval/data/splean27x8_k11_t8_seed1.out	Cuckatoo27	000005eed10000000000...0000	0-3	8	11	844.2 KiB	4
splean27x8_k11_t8	../eval/data/splean27x8_k11_t8_seed2.out	Cuckatoo27	000000000005eed200000...0000	0-3	8	11	844.2 KiB	4
splean27x8_k12_t1	../eval/data/splean27x8_k12_t1_seed0.out	Cuckatoo27	00000000000000000000...0000	0-3	1	12	427.1 KiB	4
splean27x8_k12_t1	../eval/data/splean27x8_k12_t1_seed1.out	Cuckatoo27	000005eed10000000000...0000	0-3	1	12	427.1 KiB	4
splean27x8_k12_t1	../eval/data/splean27x8_k12_t1_seed2.out	Cuckatoo27	000000000005eed200000...0000	0-3	1	12	427.1 KiB	4
splean27x8_k12_t2	../eval/data/splean27x8_k12_t2_seed0.out	Cuckatoo27	00000000000000000000...0000	0-3	2	12	428.1 KiB	4
splean27x8_k12_t2	../eval/data/splean27x8_k12_t2_seed1.out	Cuckatoo27	000005eed10000000000...0000	0-3	2	12	428.1 KiB	4
splean27x8_k12_t2	../eval/data/splean27x8_k12_t2_seed2.out	Cuckatoo27	000000000005eed200000...0000	0-3	2	12	428.1 KiB	4
splean27x8_k12_t4	../eval/data/splean27x8_k12_t4_seed0.out	Cuckatoo27	00000000000000000000...0000	0-3	4	12	428.2 KiB	4
splean27x8_k12_t4	../eval/data/splean27x8_k12_t4_seed1.out	Cuckatoo27	000005eed10000000000...0000	0-3	4	12	428.2 KiB	4
splean27x8_k12_t4	../eval/data/splean27x8_k12_t4_seed2.out	Cuckatoo27	000000000005eed200000...0000	0-3	4	12	428.2 KiB	4
splean27x8_k12_t8	../eval/data/splean27x8_k12_t8_seed0.out	Cuckatoo27	00000000000000000000...0000	0-3	8	12	428.2 KiB	4
splean27x8_k12_t8	../eval/data/splean27x8_k12_t8_seed1.out	Cuckatoo27	000005eed10000000000...0000	0-3	8	12	428.2 KiB	4
splean27x8_k12_t8	../eval/data/splean27x8_k12_t8_seed2.out	Cuckatoo27	000000000005eed200000...0000	0-3	8	12	428.2 KiB	4

[illegible]

[illegible]

[illegible]

							KiB	
splean31x8_k15_t2	../eval/data/splean31x8_k15_t2_seed0.out	Cuckatoo31	00000000000000000000...0000	0-3	2	15	940.3 KiB	4
splean31x8_k15_t2	../eval/data/splean31x8_k15_t2_seed1.out	Cuckatoo31	000005eed10000000000...0000	0-3	2	15	940.3 KiB	4
splean31x8_k15_t2	../eval/data/splean31x8_k15_t2_seed2.out	Cuckatoo31	000000000005eed200000...0000	0-3	2	15	940.3 KiB	4
splean31x8_k15_t4	../eval/data/splean31x8_k15_t4_seed0.out	Cuckatoo31	00000000000000000000...0000	0-3	4	15	940.3 KiB	4
splean31x8_k15_t4	../eval/data/splean31x8_k15_t4_seed1.out	Cuckatoo31	000005eed10000000000...0000	0-3	4	15	940.3 KiB	4
splean31x8_k15_t4	../eval/data/splean31x8_k15_t4_seed2.out	Cuckatoo31	000000000005eed200000...0000	0-3	4	15	940.3 KiB	4
splean31x8_k15_t8	../eval/data/splean31x8_k15_t8_seed0.out	Cuckatoo31	00000000000000000000...0000	0-3	8	15	940.4 KiB	4
splean31x8_k15_t8	../eval/data/splean31x8_k15_t8_seed1.out	Cuckatoo31	000005eed10000000000...0000	0-3	8	15	940.4 KiB	4
splean31x8_k15_t8	../eval/data/splean31x8_k15_t8_seed2.out	Cuckatoo31	000000000005eed200000...0000	0-3	8	15	940.4 KiB	4

Lean solver baseline calculations:

Exec. File	Output File	Variant	Seed	Nonces	Threads	Memory used	k_{mem}
lean27x8_t1	../eval/data/lean27x8_t1_seed0.out	Cuckatoo27	00000000000000000000...0000	0-3	1	32.0 MiB	0.250
lean27x8_t1	../eval/data/lean27x8_t1_seed1.out	Cuckatoo27	000005eed10000000000...0000	0-3	1	32.0 MiB	0.250
lean27x8_t1	../eval/data/lean27x8_t1_seed2.out	Cuckatoo27	000000000005eed200000...0000	0-3	1	32.0 MiB	0.250
lean27x8_t2	../eval/data/lean27x8_t2_seed0.out	Cuckatoo27	00000000000000000000...0000	0-3	2	32.0 MiB	0.250
lean27x8_t2	../eval/data/lean27x8_t2_seed1.out	Cuckatoo27	000005eed10000000000...0000	0-3	2	32.0 MiB	0.250
lean27x8_t2	../eval/data/lean27x8_t2_seed2.out	Cuckatoo27	000000000005eed200000...0000	0-3	2	32.0 MiB	0.250
lean27x8_t4	../eval/data/lean27x8_t4_seed0.out	Cuckatoo27	00000000000000000000...0000	0-3	4	32.0 MiB	0.250
lean27x8_t4	../eval/data/lean27x8_t4_seed1.out	Cuckatoo27	000005eed10000000000...0000	0-3	4	32.0 MiB	0.250
lean27x8_t4	../eval/data/lean27x8_t4_seed2.out	Cuckatoo27	000000000005eed200000...0000	0-3	4	32.0 MiB	0.250
lean27x8_t8	../eval/data/lean27x8_t8_seed0.out	Cuckatoo27	00000000000000000000...0000	0-3	8	32.0 MiB	0.250
lean27x8_t8	../eval/data/lean27x8_t8_seed1.out	Cuckatoo27	000005eed10000000000...0000	0-3	8	32.0 MiB	0.250
lean27x8_t8	../eval/data/lean27x8_t8_seed2.out	Cuckatoo27	000000000005eed200000...0000	0-3	8	32.0 MiB	0.250
lean29x8_t1	../eval/data/lean29x8_t1_seed0.out	Cuckatoo29	00000000000000000000...0000	0-3	1	128.0 MiB	0.250
lean29x8_t1	../eval/data/lean29x8_t1_seed1.out	Cuckatoo29	000005eed10000000000...0000	0-3	1	128.0 MiB	0.250
lean29x8_t1	../eval/data/lean29x8_t1_seed2.out	Cuckatoo29	000000000005eed200000...0000	0-3	1	128.0 MiB	0.250
lean29x8_t2	../eval/data/lean29x8_t2_seed0.out	Cuckatoo29	00000000000000000000...0000	0-3	2	128.0 MiB	0.250
lean29x8_t2	../eval/data/lean29x8_t2_seed1.out	Cuckatoo29	000005eed10000000000...0000	0-3	2	128.0 MiB	0.250
lean29x8_t2	../eval/data/lean29x8_t2_seed2.out	Cuckatoo29	000000000005eed200000...0000	0-3	2	128.0 MiB	0.250
lean29x8_t4	../eval/data/lean29x8_t4_seed0.out	Cuckatoo29	00000000000000000000...0000	0-3	4	128.0 MiB	0.250
lean29x8_t4	../eval/data/lean29x8_t4_seed1.out	Cuckatoo29	000005eed10000000000...0000	0-3	4	128.0 MiB	0.250
lean29x8_t4	../eval/data/lean29x8_t4_seed2.out	Cuckatoo29	000000000005eed200000...0000	0-3	4	128.0 MiB	0.250
lean29x8_t8	../eval/data/lean29x8_t8_seed0.out	Cuckatoo29	00000000000000000000...0000	0-3	8	128.0 MiB	0.250
lean29x8_t8	../eval/data/lean29x8_t8_seed1.out	Cuckatoo29	000005eed10000000000...0000	0-3	8	128.0 MiB	0.250
lean29x8_t8	../eval/data/lean29x8_t8_seed2.out	Cuckatoo29	000000000005eed200000...0000	0-3	8	128.0 MiB	0.250
lean31x8_t1	../eval/data/lean31x8_t1_seed0.out	Cuckatoo31	00000000000000000000...0000	0-3	1	512.0 MiB	0.250
lean31x8_t1	../eval/data/lean31x8_t1_seed1.out	Cuckatoo31	000005eed10000000000...0000	0-3	1	512.0 MiB	0.250
lean31x8_t1	../eval/data/lean31x8_t1_seed2.out	Cuckatoo31	000000000005eed200000...0000	0-3	1	512.0 MiB	0.250
lean31x8_t2	../eval/data/lean31x8_t2_seed0.out	Cuckatoo31	00000000000000000000...0000	0-3	2	512.0 MiB	0.250
lean31x8_t2	../eval/data/lean31x8_t2_seed1.out	Cuckatoo31	000005eed10000000000...0000	0-3	2	512.0 MiB	0.250
lean31x8_t2	../eval/data/lean31x8_t2_seed2.out	Cuckatoo31	000000000005eed200000...0000	0-3	2	512.0 MiB	0.250
lean31x8_t4	../eval/data/lean31x8_t4_seed0.out	Cuckatoo31	00000000000000000000...0000	0-3	4	512.0 MiB	0.250
lean31x8_t4	../eval/data/lean31x8_t4_seed1.out	Cuckatoo31	000005eed10000000000...0000	0-3	4	512.0 MiB	0.250
lean31x8_t4	../eval/data/lean31x8_t4_seed2.out	Cuckatoo31	000000000005eed200000...0000	0-3	4	512.0 MiB	0.250
lean31x8_t8	../eval/data/lean31x8_t8_seed0.out	Cuckatoo31	00000000000000000000...0000	0-3	8	512.0 MiB	0.250
lean31x8_t8	../eval/data/lean31x8_t8_seed1.out	Cuckatoo31	000005eed10000000000...0000	0-3	8	512.0 MiB	0.250
lean31x8_t8	../eval/data/lean31x8_t8_seed2.out	Cuckatoo31	000000000005eed200000...0000	0-3	8	512.0 MiB	0.250

How to create and run all binary files to reproduce the full evaluation

The command

```
make all
```

can be used to compile and link all binary files with the parameter used for the evaluation. Followed by the command

```
make eval_all
```

to run these binaries and gather the output in the [data directory](#). [Jupyter Notebook](#) files in the [image directory](#) can be run to create the overview graphs and images used in the presentation of the evaluation.

How to create and run a quick test

Use the command

```
make quick_test
```

to create the files for a quick test and use

```
make run_quick_test
```

to run the test. The output will not be verbose and results will be piped to the console.

How to create and run only a part of the evaluation

However, running the full evaluation takes quite some time. Therefore, one can create and run only a part of the evaluation by using the following commands. The format of the created executable files is

(type)(bits)x(parallel hashes)_k(reduction factor)

so for example: splean31x8_k10, will execute splean solver approach for Cuckatoo31 using SIMD with 8 Siphash calculations in parallel and a k-factor of 10 (so using only a fraction $\frac{1}{2^{10}}$ of the edges to start with).

To only create all lean solver baseline binary files use

```
make baseline
```

To make lean solver baseline binaries for a certain Cuckatoo variant with a specified amount of parallel Siphash calculations, use for example (for lean solver targeting Cuckatoo27 with 4 Siphashes in parallel)

```
make lean27x4
```

To run the baseline calculations, use

```
make eval_bl
```

To create all splean solver variants, use

```
make splean
```

for a particular solver variant, use for example (for splean solver targeting Cuckatoo29 with 8 Siphash calculations in parallel):

```
make splean27x8
```

To run the evaluations for a part of the splean solver variants, use for example (for splean solver targeting Cuckatoo29 with 8 Siphash calculations in parallel):

```
make eval_splean27x8
```

To create the cuda variants of mean solver for the validation process, use:

```
make verify
```

or more specific for example (for mean solver targeting Cuckatoo27 using cuda):

```
make mean27cu
```

to run the validation to find all nonces that lead to 42-cycles, use

```
make run_verify
```

or more specific for example (for mean solver targeting Cuckatoo27 using cuda):

```
make run_verify_27
```